

Quasimatrices, Gram-Schmidt, and QR

Idea: $f \in H \iff f = c_1 e_1 + c_2 e_2 + c_3 e_3 + \dots$

Gram-Schmidt: Given $\{v_1, v_2, \dots, v_n\}$, produce an orthonormal basis for $V_n = \text{span}\{v_1, v_2, \dots, v_n\}$.

$$\Rightarrow \text{Set } \tilde{q}_1 = v_1 \text{ and } q_1 = \tilde{q}_1 / \|\tilde{q}_1\|$$

$$\Rightarrow \tilde{q}_2 = v_2 - \langle v_2, q_1 \rangle q_1 \text{ and } q_2 = \tilde{q}_2 / \|\tilde{q}_2\|$$

$\vdots$

$$\Rightarrow \tilde{q}_n = v_n - \sum_{j=1}^{n-1} \langle v_n, q_j \rangle q_j \text{ and } q_n = \tilde{q}_n / \|\tilde{q}_n\|$$

Example: Construct an $L^2(-1,1)$ -ONB for

$$P_n = \text{span}\{1, x, x^2, \dots, x^n\}.$$

$$\underline{v_1 = 1} \Rightarrow \|\tilde{q}_1\|_{L^2}^2 = \int_{-1}^{+1} 1^2 dx = 2$$

$$\Rightarrow q_1 = \tilde{q}_1 / \|\tilde{q}_1\| = 1/\sqrt{2}$$

$$\underline{v_2 = x} \Rightarrow \tilde{q}_2 = v_2 - \langle v_2, q_1 \rangle q_1$$

$$= x - \left[\int_{-1}^{+1} x \left(\frac{1}{\sqrt{2}} \right) dx \right] \left(\frac{1}{\sqrt{2}} \right)$$

$$= 0$$

$$= x$$

$$\Rightarrow \|\tilde{q}_2\|_{L^2}^2 = \int_{-1}^{+1} x^2 dx = \frac{x^3}{3} \Big|_{-1}^{+1} = \frac{2}{3}$$

$$\Rightarrow q_2 = \tilde{q}_2 / \|\tilde{q}_2\| = \sqrt{\frac{3}{2}} x$$

$$\underline{v_3 = x^2} \Rightarrow \tilde{q}_3 = v_3 - \langle v_3, q_2 \rangle q_2 - \langle v_3, q_1 \rangle q_1$$

$$\Rightarrow \langle v_3, q_2 \rangle = \int_{-1}^{+1} x^2 \left(\sqrt{\frac{3}{2}} x \right) dx = 0$$

$$\Rightarrow \langle v_3, q_1 \rangle = \int_{-1}^{+1} x^2 \left(\frac{1}{\sqrt{2}} \right) dx = \frac{1}{\sqrt{2}} \frac{2}{3} = \frac{\sqrt{2}}{3}$$

$$\Rightarrow \tilde{q}_3 = x^2 - 0 \cdot \frac{\sqrt{3}}{2} x - \frac{\sqrt{2}}{3} \frac{1}{\sqrt{2}} = x^2 - \frac{1}{3}$$

$$\Rightarrow \|\tilde{q}_3\|^2 = \int_{-1}^{+1} (x^2 - \frac{1}{3})^2 dx = \frac{8}{45}$$

$$q_3 = \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3} \right)$$

For $n=3$, $1/\sqrt{2}, \sqrt{3}/2 x, \sqrt{45}/9 (x^2 - 1/3)$

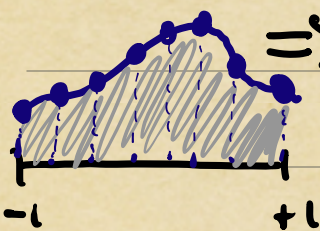
In general, applying the Gram-Schmidt algorithm to monomials $1, x, x^2, x^3, \dots$ leads to the **Legendre polynomials**, normalized to have $L^2(-1,1)$ norm equal to one.

$$l_0(x), l_1(x), l_2(x), \dots$$

Classically, the normalization is $\hat{l}_j(x) = c_j l_j(x)$ where c_j is chosen s.t. $\hat{l}_j(1) = 1$.

$\Rightarrow$ Solutions to classical ODE/PDE

$\Rightarrow$ Roots correspond to Gauss-quadrature nodes.



$\Rightarrow$ For numerical methods for PDE, for integral equations, and more

$\Rightarrow$ Apparently, also used for feature-embeddings in ML

Question: How can we use Gram-Schmidt to solve the "best approximation" problem?

E.g.) $\min_{g \in V_n} \|f - g\|$ where $V_n \subset H$.

A quasimatrix w/ columns $e_1, \dots, e_n \in H$:

$$E = \begin{bmatrix} 1 & & 1 \\ e_1 & \dots & e_n \\ 1 & & 1 \end{bmatrix} \quad \begin{pmatrix} \Omega \\ [-1, 1] \times n \\ \infty \times n \end{pmatrix}$$

Linear Combs: $\underline{c} = (c_1, \dots, c_n)^T \in \mathbb{C}^n$

$$E\underline{c} = \begin{bmatrix} 1 & & 1 \\ e_1 & \dots & e_n \\ 1 & & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = c_1 e_1 + \dots + c_n e_n \in H$$

$E: \mathbb{C}^n \rightarrow H$ linear

Inner Products: $f \in H$ $E^*: H \rightarrow \mathbb{C}^n$ linear

$$E^* f = \begin{bmatrix} - & e_1^* & - \\ \vdots & \vdots & \vdots \\ - & e_n^* & - \end{bmatrix} \begin{bmatrix} 1 \\ f \\ 1 \end{bmatrix} = \begin{bmatrix} \langle f, e_1 \rangle \\ \vdots \\ \langle f, e_n \rangle \end{bmatrix} \in \mathbb{C}^n$$

Gram Matrix: $G = E^* E \in \mathbb{C}^{n \times n}$

$$G = \begin{bmatrix} -e_1^* - \\ \vdots \\ -e_n^* - \end{bmatrix} \begin{bmatrix} | & | \\ e_1 & \dots & e_n \\ | & | \end{bmatrix} = \begin{bmatrix} \langle e_1, e_1 \rangle & \dots & \langle e_n, e_1 \rangle \\ \vdots & & \vdots \\ \langle e_1, e_n \rangle & \dots & \langle e_n, e_n \rangle \end{bmatrix}$$

Q: What if $\{e_1, \dots, e_n\}$ are orthonormal?

$$G = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} = I$$

"Orthogonal":
Projection:

$$P = E E^* : H \rightarrow H$$

$\hat{=}$ " $e_1, \dots, e_n$ " are orthonormal

Take $f \in H$,

$$Pf = \begin{bmatrix} | & | \\ e_1 & \dots & e_n \\ | & | \end{bmatrix} \begin{bmatrix} -e_1^* - \\ \vdots \\ -e_n^* - \end{bmatrix} \begin{bmatrix} | \\ f \\ | \end{bmatrix} = \begin{bmatrix} | & | \\ e_1 & \dots & e_n \\ | & | \end{bmatrix} \begin{bmatrix} \langle f, e_1 \rangle \\ \vdots \\ \langle f, e_n \rangle \end{bmatrix}$$

$$= \langle f, e_1 \rangle e_1 + \dots + \langle f, e_n \rangle e_n$$

$$= \sum_{j=1}^n \langle f, e_j \rangle e_j$$

"Outer Product": e_1 and $e_2 \in H$, $e, e^*: H \rightarrow H$

$$e, e_2^* \xi = \begin{bmatrix} 1 \\ e_1 \\ 1 \end{bmatrix} [-e_2^* -] \begin{bmatrix} 1 \\ \xi \\ 1 \end{bmatrix} = \langle \xi, e_2 \rangle e_1$$

The QR Decomposition

Given a $n \times n$ matrix w/lin. indep. cols,

$$E = \begin{bmatrix} 1 & & 1 \\ e_1 & \dots & e_n \\ 1 & & 1 \end{bmatrix},$$

Gram-Schmidt finds an ONB for

$$\text{col}(E) = \text{span}\{e_1, e_2, \dots, e_n\},$$

by taking linear combos of $e_1, \dots, e_n$.

$$\begin{bmatrix} 1 & & 1 \\ e_1 & \dots & e_n \\ 1 & & 1 \end{bmatrix} \begin{bmatrix} x & x & \dots & x \\ 0 & x & \dots & x \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} = \begin{bmatrix} 1 & & 1 \\ q_1 & \dots & q_n \\ 1 & & 1 \end{bmatrix}$$

$E \qquad T \qquad Q$

$$\tilde{q}_j = \frac{v_j - \sum_{k=1}^{j-1} \langle v_j, e_k \rangle e_k}{\|\tilde{q}_j\|} \xrightarrow{\text{Gram-Schmidt}} Q$$

$$\begin{bmatrix} 1 & & \\ e_1 & \dots & e_n \\ 1 & & \end{bmatrix} = \begin{bmatrix} 1 & & \\ q_1 & \dots & q_n \\ 1 & & \end{bmatrix} \begin{bmatrix} x & \dots & x \\ 0 & x & \dots & x \\ 0 & 0 & \dots & x \\ 0 & 0 & \dots & 0 & x \end{bmatrix}$$

E
 Q
 $T^{-1} = R$